

Modeling Discrete Linear Change and Arithmetic Sums

Building arithmetic models, solving for the term number, and adding up a finite run of terms

Directions

- Each situation changes by the same amount at every stage, so it is an arithmetic sequence. State what n counts before you compute anything.
 - Reaching term n from term 1 takes $n - 1$ steps: $a_n = a_1 + (n - 1)d$.
 - A total over the first n stages is $S_n = \frac{n}{2}(a_1 + a_n)$ — pair the smallest with the largest.
 - Some problems ask for a value, some for the stage number, and some for a running total. Read which one before choosing a formula. Keep units with your answers.
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1. A single stacking chair stands 82 cm tall. Each chair added to the stack raises it by 9 cm. How tall is a stack of 6 chairs?

Answer: _____ cm

2. A gym charges a \$30 joining fee plus \$22 for each month of membership. How much has a member paid in total after 8 months, in dollars?

Answer: _____

3. A runner trains 12 minutes in week 1 and adds 4 minutes in each week after that. How many minutes does she run in total over the first 8 weeks?

Answer: _____ min

4. A water tank holds 240 litres and loses 15 litres each day, so after n days it holds $240 - 15n$ litres. On which day is the tank empty?

Answer: _____

5. A theatre has 20 seats in row 1 and 3 more seats in each row after that. How many seats are in row 15?

Answer: _____

6. The same theatre has 20 seats in row 1 and 3 more in each row after that. How many seats are there altogether in rows 1 through 15?

Answer: _____

7. A savings account holds \$45 after week 1, and \$18 is added every week after that. In which week does the balance first reach \$261?

Answer: _____

8. An emergency plumber charges \$95 for the first hour on site and \$60 for each additional hour. What does a 7-hour call-out cost, in dollars?

Answer: _____

9. A brick pile has 48 bricks in the bottom row, and each row above holds 3 fewer than the row below it. The pile is 12 rows high. How many bricks are in the pile altogether?

Answer: _____

10. A phone plan gives 500 MB of data in month 1 and 250 MB more in each month after that. In which month does the allowance first reach 3000 MB?

Answer: _____

11. Two job offers both raise pay by a fixed amount each year. Offer *A* pays \$2400 in year 1 and \$1500 more in each year after that. Offer *B* pays \$3600 in year 1 and \$900 more in each year after that.

(a) In which year do the two offers pay the same? year = _____

(b) What does offer *A* pay in total over its first 6 years, in dollars? total = _____

12. A concert hall has 24 rows. Row 1 holds 18 seats and each row behind it holds 2 more.

(a) How many seats are in the last row? seats = _____

(b) How many seats does the hall hold altogether? total = _____

(c) The manager says that building 24 more rows in the same pattern would double the hall's capacity. Explain why the capacity would in fact more than double.